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Python Test Questions

1. Conditions (if, elif, else)

Q1. Write a program to check whether a number is positive, negative, or zero.

```
>>> num = int(input("Enter a number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

Q2. Write a program to find the largest of three numbers.

```
>>> a = int(input())
b = int(input())
c = int(input())

if a >= b and a >= c:
    print(a)
elif b >= a and b >= c:
    print(b)
else:
    print(c)
```

Q3. Write a program to check whether a given year is a leap year.

```
>>> year = int(input("Enter a year: "))

if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print(year, "is a leap year.")
else:
    print(year, "is not a leap year.")
```

Q4. Write a program to check if a character is a vowel or consonant.

```
>>> ch = input("Enter a character: ")

if ch in "aeiouAEIOU":
    print("Vowel")
else:
    print("Consonant")
```

Q5. Write a program to calculate the electricity bill:

- First 100 units → ₹5/unit
- Next 100 units → ₹7/unit
- Above 200 units → ₹10/unit

```
>>>
```

?

2. Loops (for / while)

Q1. Print all even numbers from 1 to 100.

>>>

```
for i in range(2, 101, 2):  
    print(i)
```

Q2. Find the factorial of a given number using a loop.

>>>

```
n = int(input("Enter a number: "))  
fact = 1
```

```
for i in range(1, n + 1):  
    fact *= i  
print("Factorial =", fact)
```

Q3. Print the Fibonacci series up to n terms

>>>

Q4. Find the sum of digits of a given number.

>>>

```
n = int(input("Enter a number: "))  
s = 0
```

```
while n > 0:  
    s += n % 10  
    n //= 10
```

```
print("sum =", s)
```

Q5. Check whether a number is an Armstrong number.

>>>

3. Functions

Q1. Write a function to find the greatest of three numbers.

>>>

```
def gratest(a, b, c):  
    return max(a, b, c)
```

```
a = int(input())  
b = int(input())  
c = int(input())
```

```
print("Greatest =", gratest(a, b, c))
```

Q2. Write a function to check whether a number is prime.

>>>

```
def prime(n):  
    if n <= 1:  
        return False
```

```

    for i in range(2, n):
        if n % i == 0:
            return False
    return True

```

```

n = int(input())

```

```

if prime(n):
    print("Prime")
else:
    print("Not Prime")

```

Commented [U1]:

Commented [U2R1]:

Q3. Write a function that returns the reverse of a string.

```

>>>
def reverse(s):
    return s[::-1]

```

```

text = input()

```

```

print(reverse(text))

```

Q4. Write a function to calculate the area and perimeter of a rectangle.

```

>>>
def rectangle(length, width):
    area = length * width
    perimeter = 2 * (length + width)
    return area, perimeter

```

```

l = float(input("Enter length: "))
w = float(input("Enter width: "))

```

```

a, p = rectangle(l, w)

```

```

print("Area =", a)
print("perimeter =", p)

```

Q5. Write a function that accepts a list and returns the second largest element.

```

>>>

```

4. Exception Handling

Q1. Write a program to handle ZeroDivisionError.

```

>>>
try:
    a = int(input())
    b = int(input())
    print(a / b)
except ZeroDivisionError:
    print("Cannot divide by zero")

```

Q2. Write a program to handle ValueError when the user enters invalid input.

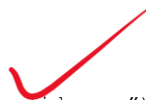
```
>>>
try:
    n = int(input())
    print(n)
except ValueError:
    print("Invalid input")
```



Q3. Create a custom exception to check if age is less than 18.

```
>>>
class AgeError(Exception):
    pass

try:
    age = int(input())
    if age < 18:
        raise AgeError("Invalid Age! Enter valid age.")
    print("Eligible")
except AgeError:
    print("Age must be 18 or above")
```



Q4. Write a program that handles both IndexError and KeyError.

```
>>>
```



Q5. Write a program using try, except, else, and finally to read a file and display its contents.

```
>>>
```